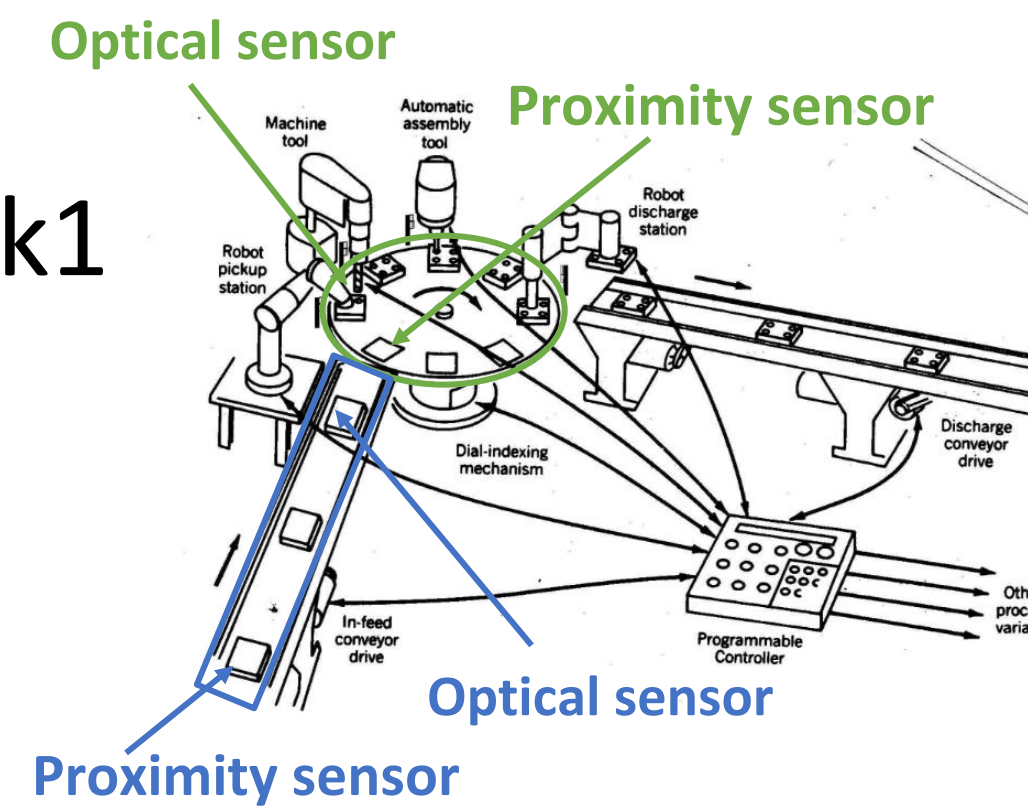
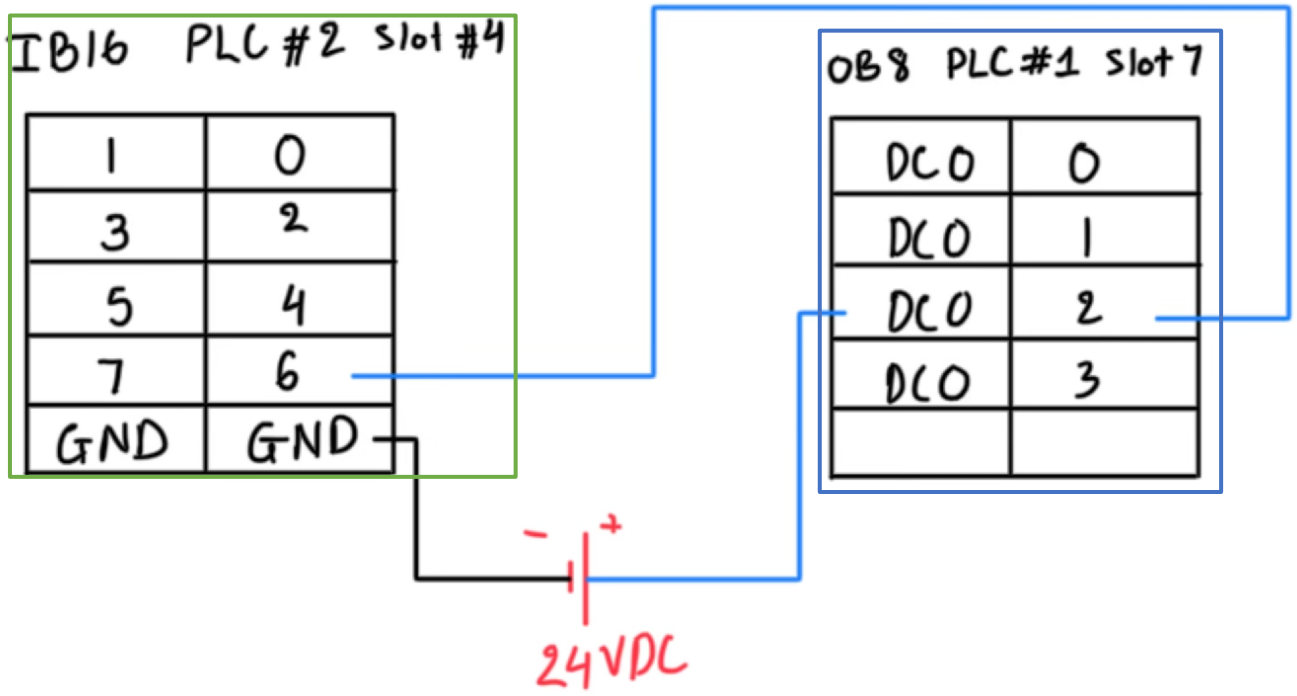
# MMET 410 Lab 9

# Task1

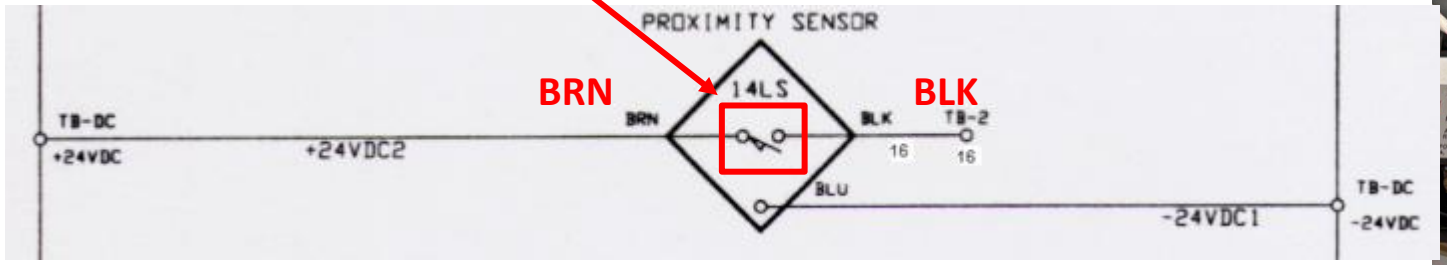


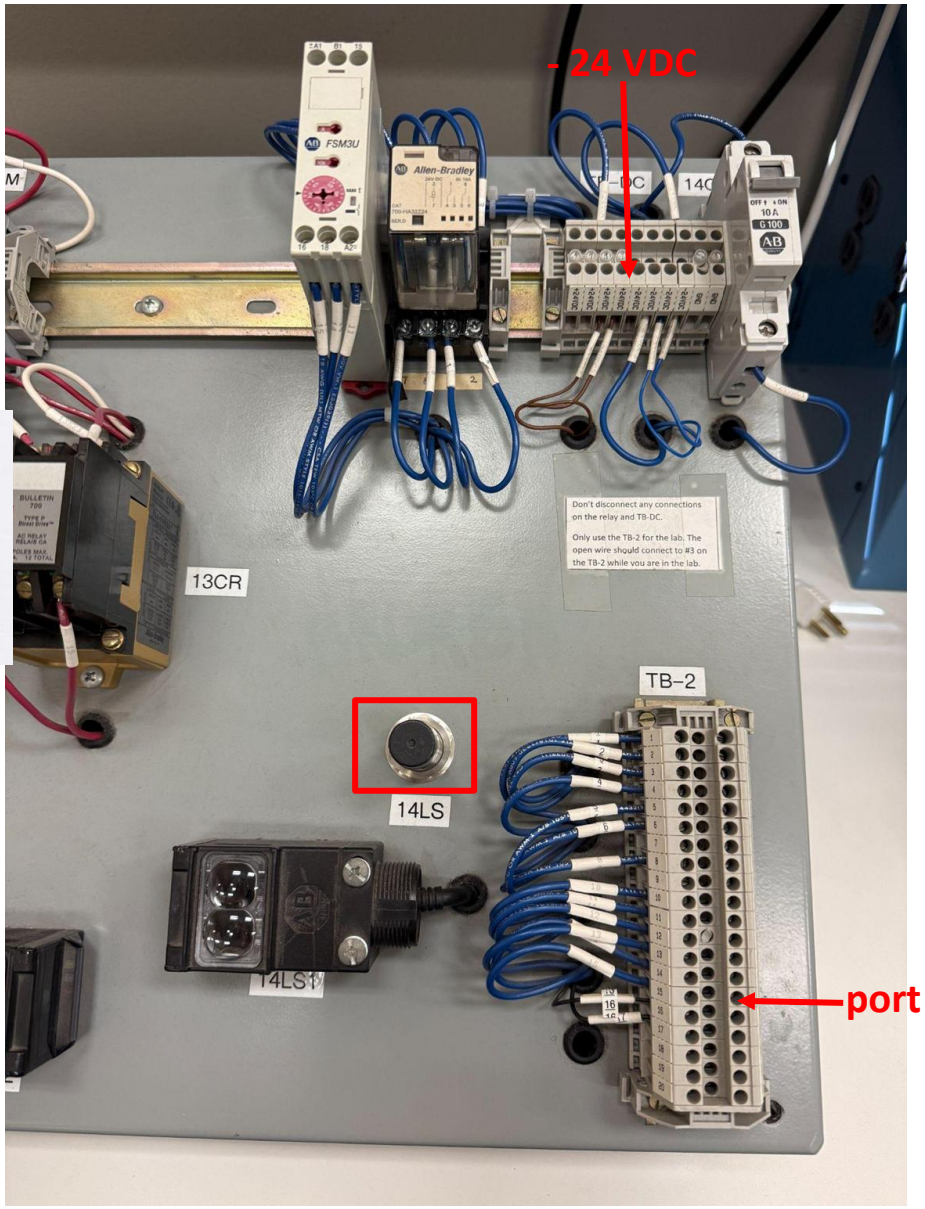




1. Team up in a group of two.
2. Work with your neighbor group.
3. 1 group work on PLC1 (Conveyor), 1 group work on PLC2 (Index table).
4. Draw the ladder diagram.
5. Wire up the connections.

# Task 2 Test the functionality of the proximity sensor.

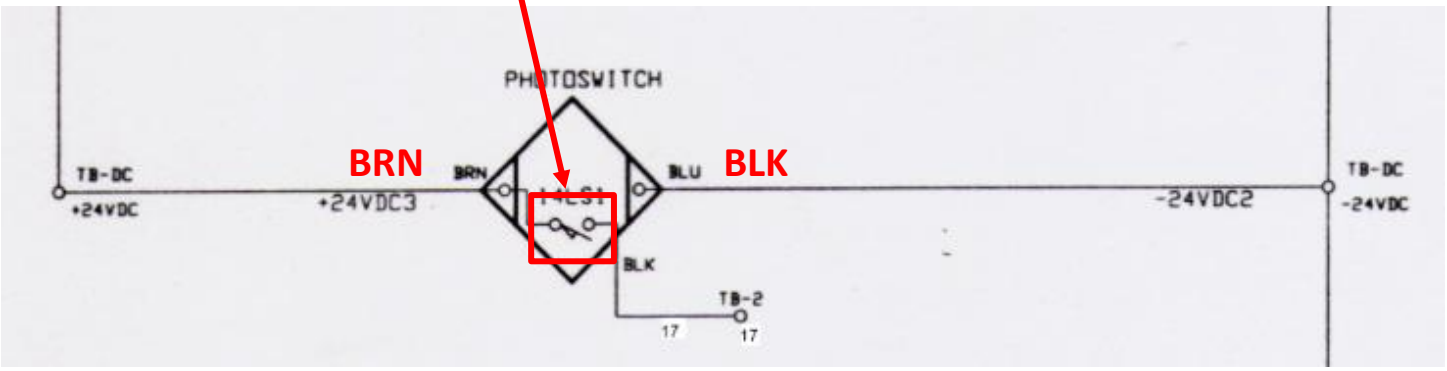
Normally open switch

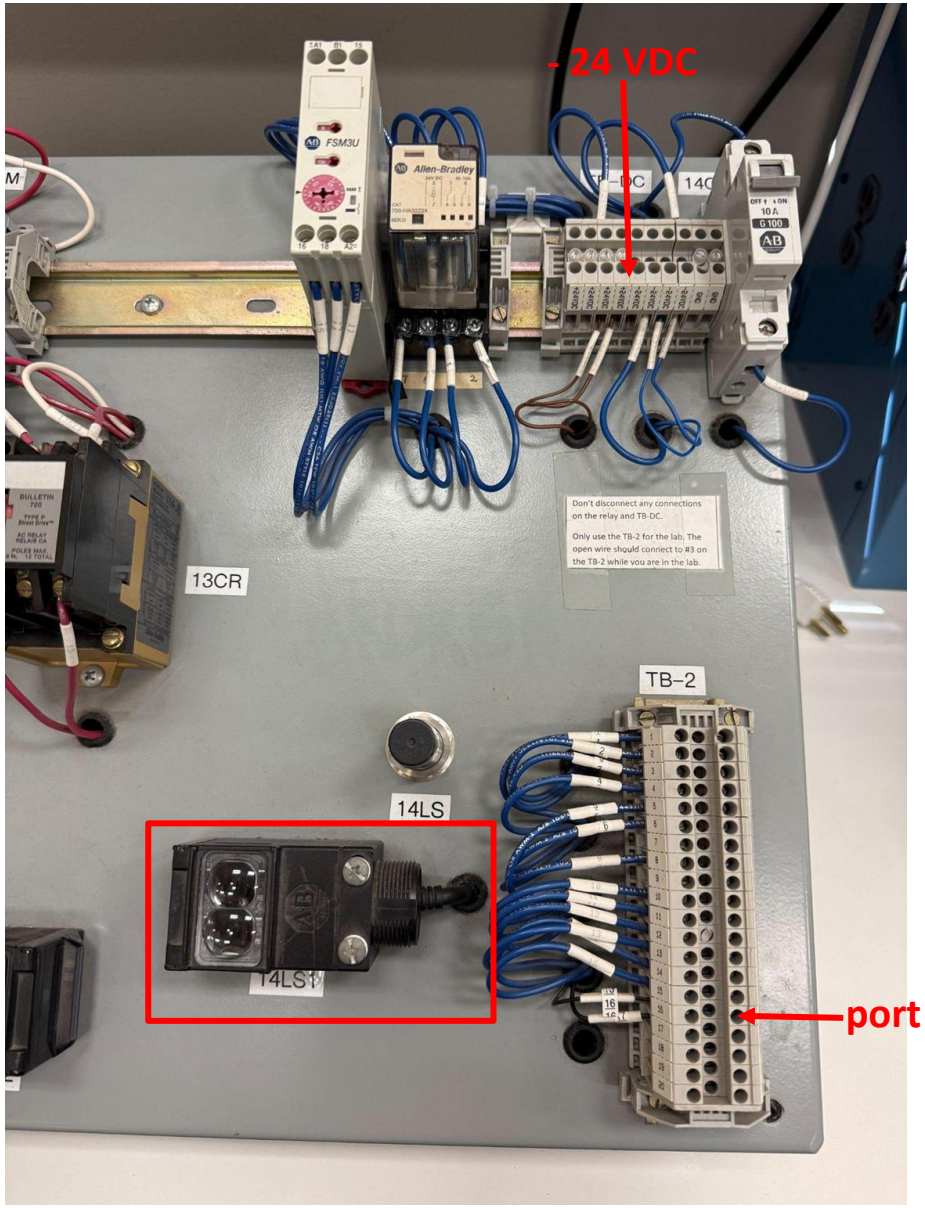


# Task 3 Test the functionality of an optical sensor.

Normally open switch



# Task 2 Test the functionality of the proximity sensor.

- c. Examine the schematic drawing in Figure 5. Identify normally open (N.O.) terminals: **Brown and Black terminals**
  Check voltage reading between TB2-16 and TB -24VDC: **0 Volts**
- d. Place a piece of metal, such as a screwdriver blade, on the top of the proximity sensor. Check voltage reading between TB2-16 and TB– **24VDC: 23.25 V**
- Summarize: What can you conclude from the voltage readings? **The proximity sensor closes the circuit when a piece of metal comes in contact with it.**

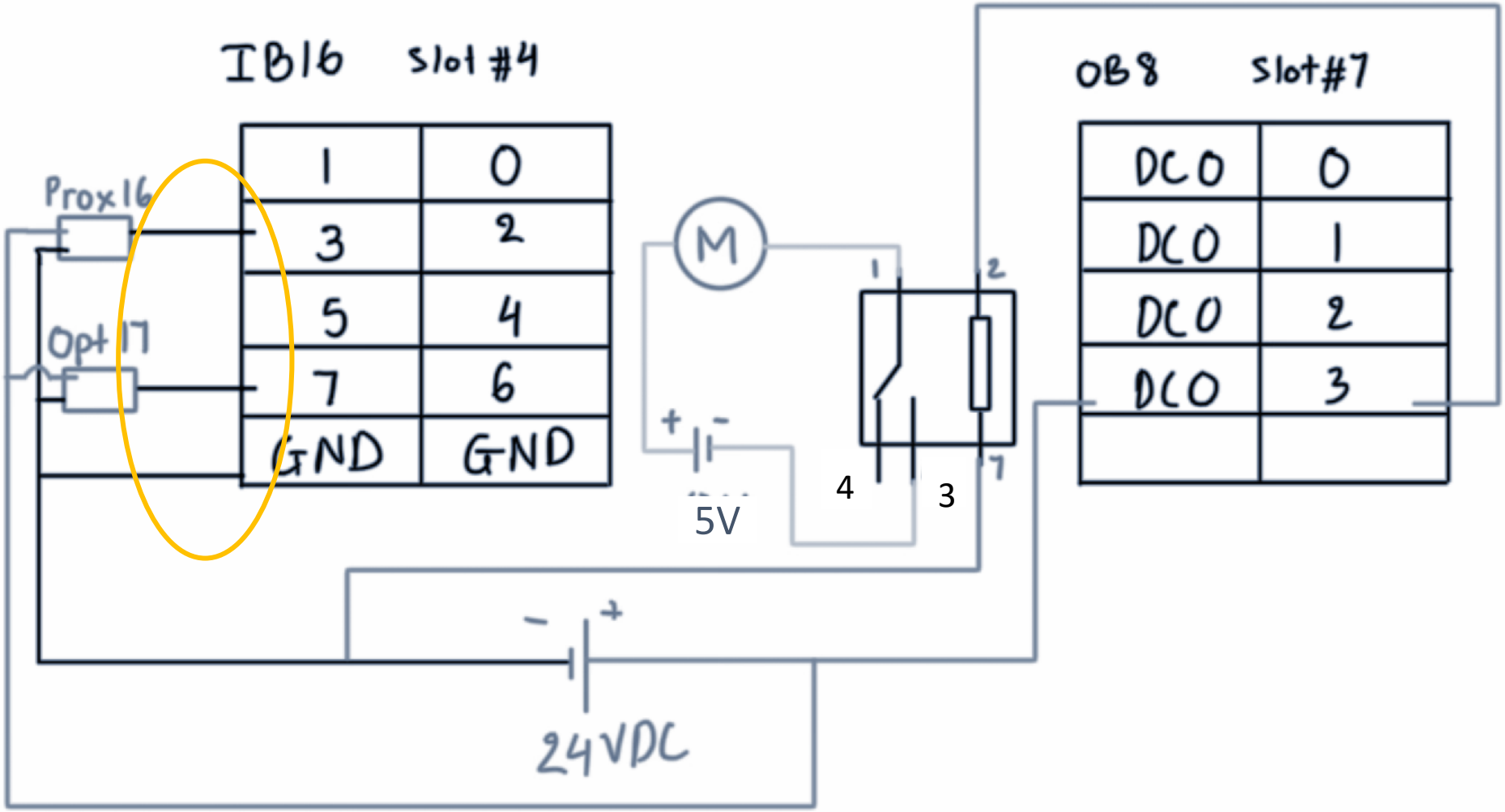
# Task 3 Test the functionality of an optical sensor.

- a. Examine the schematic drawing in Figure 5. Identify normally open (N.O.) terminals:
  **Brown and Black terminals**
- b. Place your fingers on top of the photo-switch sensor until all indicators on top of the sensor are off. Check voltage reading between TB2-17 and TB -24VDC: **23.18 V**
- c. Place some flat shiny reflector material above the photo-switch. Check voltage reading between TB2-17 and TB–24VDC
  **It varies depending on how the light is shining or reflecting, the angle of the shiny object also affects it. It read about 0 volts with a certain angle, then the reading went back to 23.19 volts when the angle was changed.**
- Summarize: What can you conclude from the voltage readings?
  **The optical sensor closes contacts when an object is detected in front of it. It also uses light reflection to detect the object, so shiner objects can cause it to miss an object at certain angles.**

# Task 4 and 5

Test the DC input modules using **proximity** and **optical** sensors.

Test the **relay** to drive the **motor**.



| Clarify | |
|---|---|
| Power supply 5v | Motor |
| PLC | Prox/opt sensor |
| Ground in PLC | -24 |
| DCO in PLC | +24 |

Prox/opt sensor: only one port need to be connected.



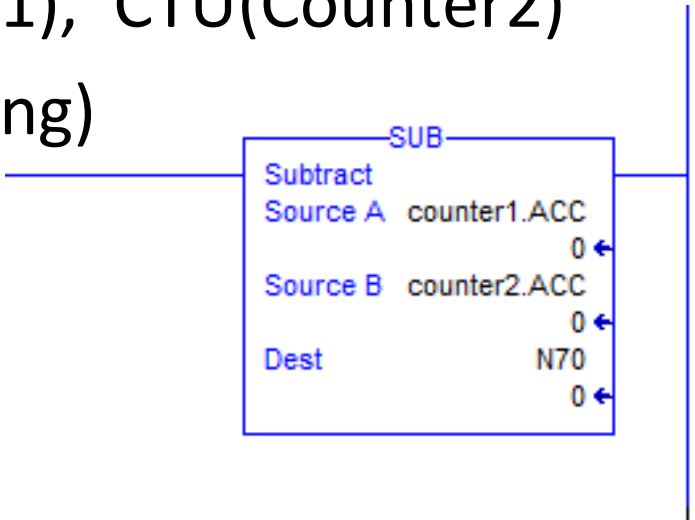

Use the GND-0 (The top one)

# Task 6

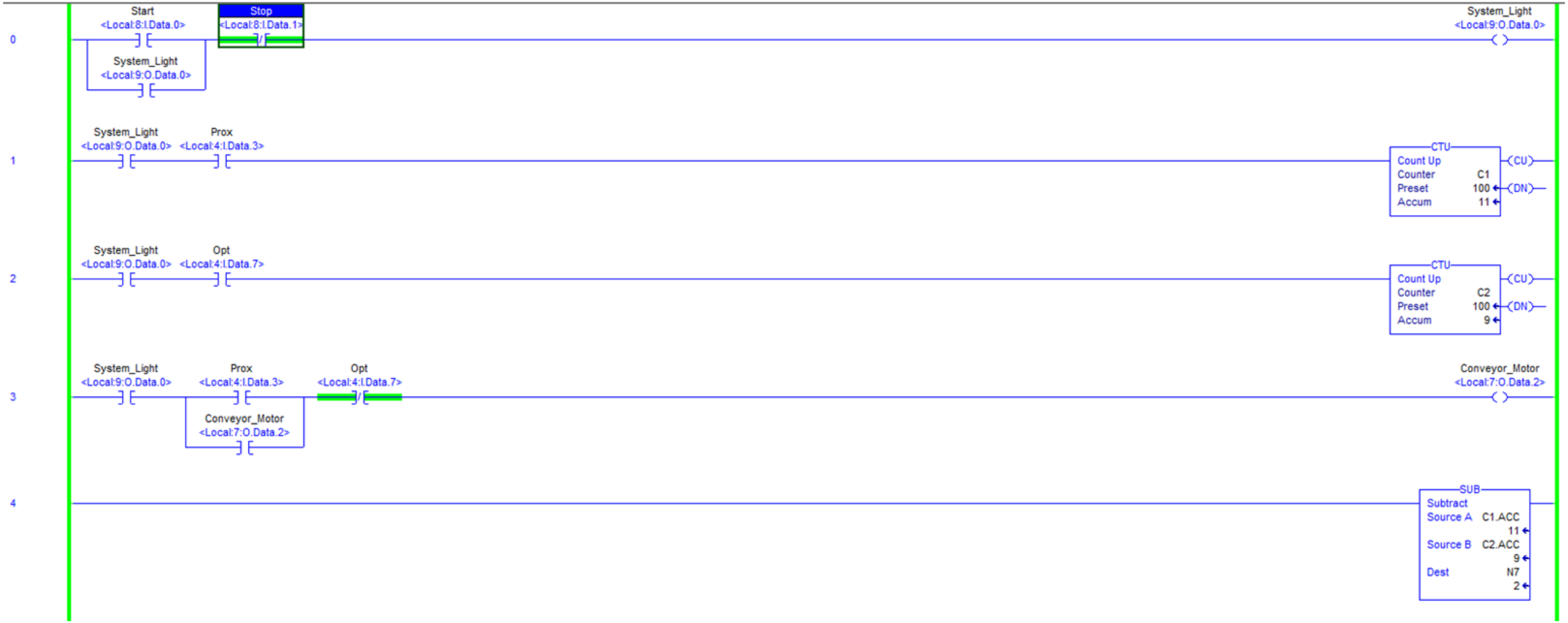
- Procedure:
- start → system on → **(proximity sensor triggered → counter 1 added 1 → conveyor on → optical sensor triggered → counter 2 added 1 → conveyor off ) (can be repeated)** → stop – all shut down
- Use SUB in last rung

- **4 input**
  Start(Slot 8, IN-0) Stop(Slot 8, IN-1)
  Proximity sensor(Slot 4, IN-3)
  Optical sensor(Slot 4, IN-7)
- **2 output**
  System(Slot 9, OUT-0)
  Conveyor(Slot 7, OUT-3)

- **2 counter**
  CTU(Counter1), CTU(Counter2)
  SUB(Remaining)



# Task6


0
Start
<Local:8:I.Data.0>
Stop
<Local:8:I.Data.1>
System_Light
<Local:9:O.Data.0>
System_Light
<Local:9:O.Data.0>
1
System_Light
<Local:9:O.Data.0>
Prox
<Local:4:I.Data.3>
CTU
Count Up
Counter C1
Preset 100
Accum 11
CU
DN
2
System_Light
<Local:9:O.Data.0>
Opt
<Local:4:I.Data.7>
CTU
Count Up
Counter C2
Preset 100
Accum 9
CU
DN
3
System_Light
<Local:9:O.Data.0>
Prox
<Local:4:I.Data.3>
Opt
<Local:4:I.Data.7>
Conveyor_Motor
<Local:7:O.Data.2>
Conveyor_Motor
<Local:7:O.Data.2>
4
SUB
Subtract
Source A C1.ACC
11
Source B C2.ACC
9
Dest N7
2